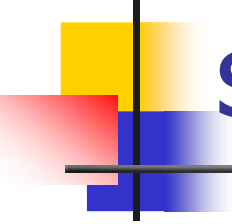


Satellite Subsystems



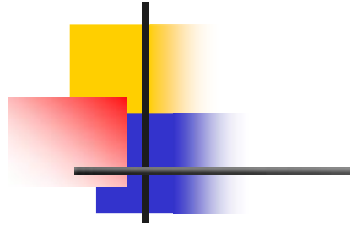
Irrespective of the intended application, be it a communications satellite or a weather forecasting satellite or even a remote sensing satellite, different subsystems comprising a typical satellite include the following:

1. Mechanical structure
2. Propulsion subsystem
3. Thermal control subsystem
4. Power supply subsystem
5. Telemetry, tracking and command (TT&C) subsystem
6. Attitude and orbit control subsystem (AOCS)
7. Payload subsystem/Communication subsystem
8. Antenna subsystem



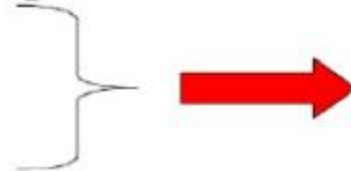
Important subsystems

- 1. Altitude and orbit Control System (AOCS)
- 2. Telemetry, Tracking, Command and Monitoring (TT&M)
- 3. Power System
- 4. Communication Subsystem
- 5. Satellite Antennas



1. AOCS (Attitude & orbit control system)

- AOCS is needed to get the satellite into the correct orbit and keep it there
 - Orbit insertion
 - Orbit maintenance
 - Fine pointing
- Major parts
 - Attitude control system
 - Orbit control system

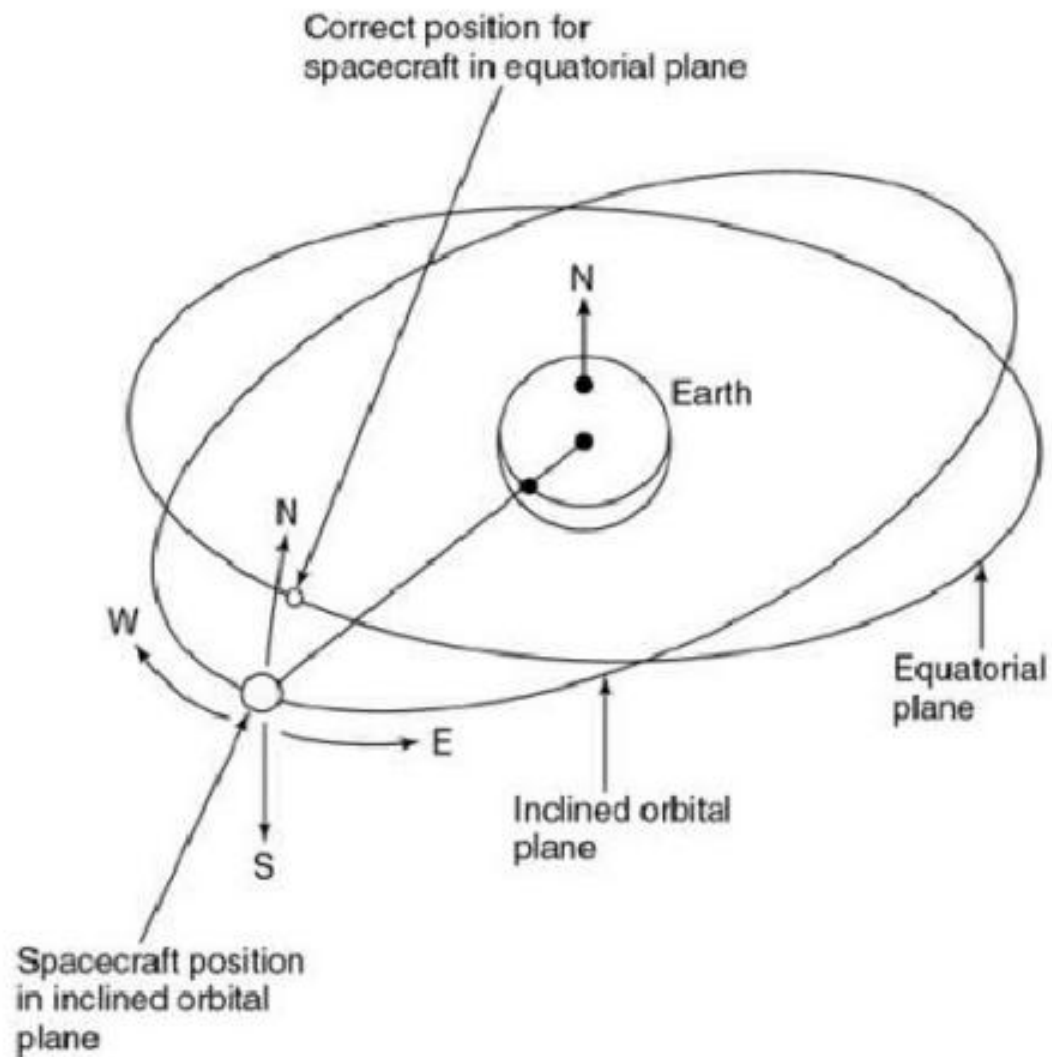


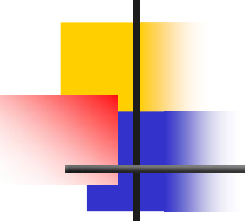


Reason for Attitude & Orbit control

- At GEO orbit altitude the moon's gravitational force is about twice as strong as the sun's
 - Moon orbit is inclined to the equatorial plane by approximately 5 degrees
 - The plane of the earth's rotation around the sun is inclined to 23 degrees to the equatorial plane
- Net gravitational force on the satellite tends to change the inclination of the satellite.
- Also solar pressure acting on solar cells and antennas of the satellite and eddy currents generated by earth's magnetic field cause rotation in the satellite. Orbital period of the satellite makes many effects cyclic, which can cause "Nutation" (a wobble) of the satellite.

Inclined Orbit

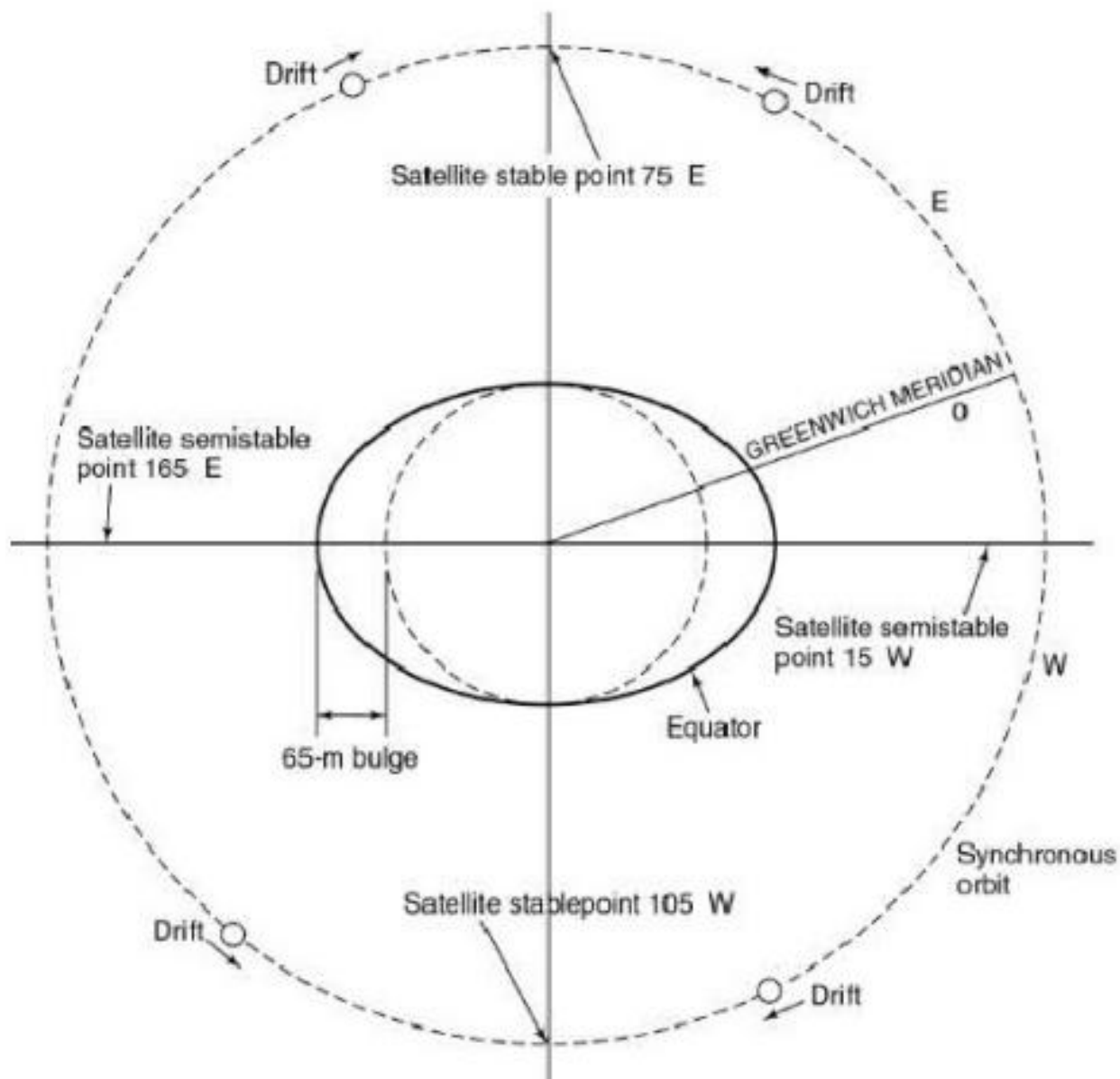
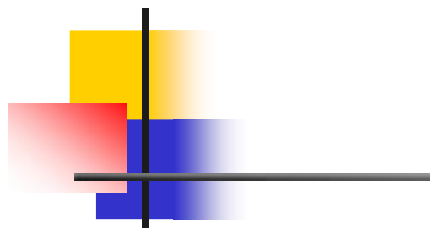


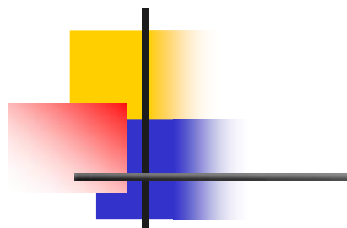


Reason for Orbit Correction (Contd..)

- Inclination rate is approximately 0.86 degrees per year from the equatorial plane.
- LEO satellites are less effected by this gravitational pull from the sun and moon
- At the equator there are bulges of about 65m at longitudes 162 degrees East and 348 degrees East.

Thus Satellite is accelerated towards one of two stable points on GEO orbit at the longitude of 75 degree E and 252 degrees E



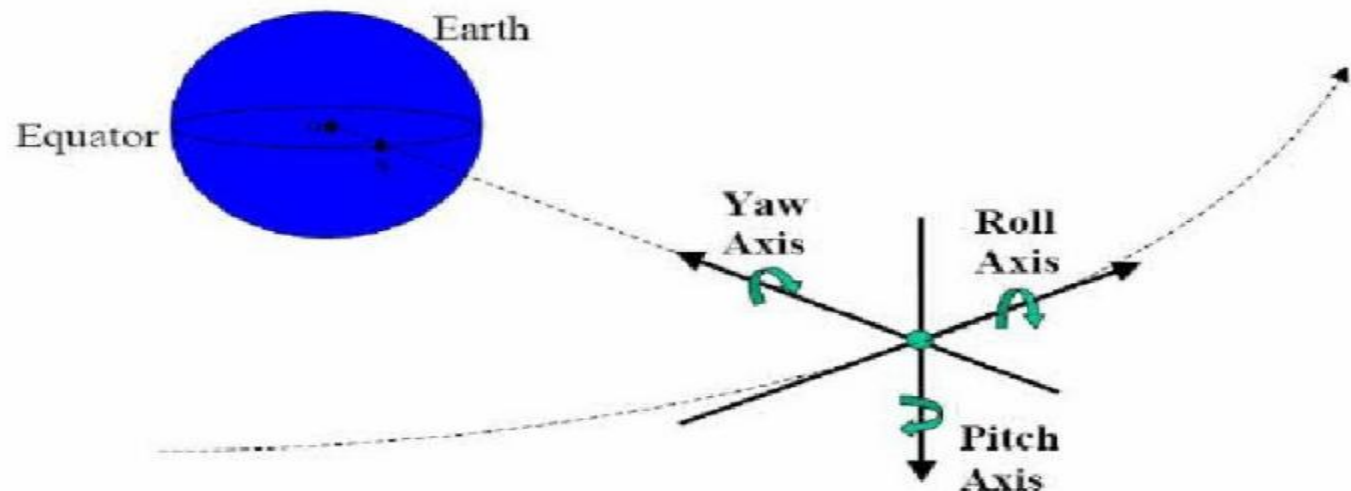


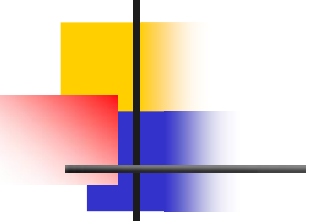
Fine positioning

- Two ways to make the satellite stable in orbit when it is weightless.
 - Satellite can be rotated at a rate between 30 and 100 rpm to create gyroscopic force that provides stability (spinner satellites)
 - Satellites can be stabilized by one or more momentum wheels, called three-axis stabilized satellites.

Definition of axis

- Roll axis
 - Rotates around the axis tangent to the orbital plane (*N/S* on the earth)
- Pitch axis
 - Moves around the axis perpendicular to the orbital plane (*E/W* on the earth)
- Yaw axis
 - Moves around the axis of the sub-satellite point

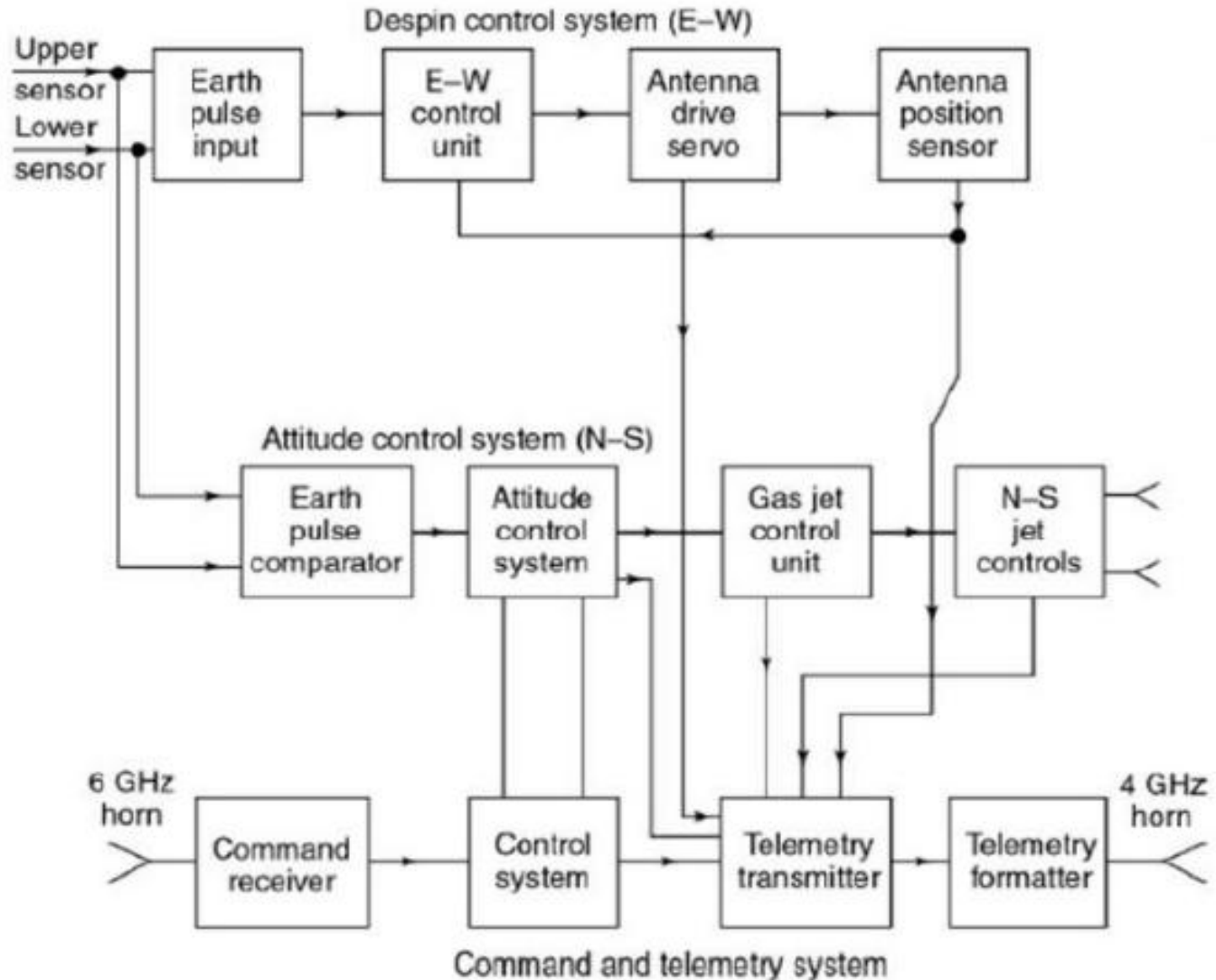


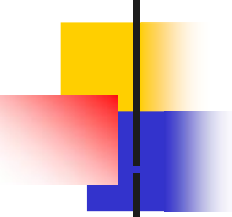


Orbit insertion & Maintenance- GEO

- Two types of motors used on satellites.
 - Traditional bipropellant thruster
 - Bipropellants used are Mono-methyl Hydrazine and Nitrogen tetroxide
 - They are hypogolic, i.e., they ignite simultaneously on contact without any catalyst or heater
 - Arc jets or ion thrusters
 - High voltage is used to accelerate ions
- Fuel stored in GEO satellite is used for two purposes
 - Apogee kick motor (AKM) that injects the satellite into its final orbit
 - Maintain the satellite in that orbit over its lifetime.

On Board Control System





2. TTC&M

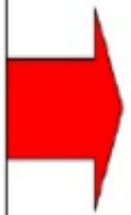
- Major functions
 - Report spacecraft health
 - Monitor command actions
 - Determine orbital elements
 - Launch sequence deployment
 - Control of thrusters
 - Control of payload (communications, etc.)

TTC&M is often a battle between *Operations* (who want every little thing monitored) and *Engineering* who want to hold data channels to a minimum

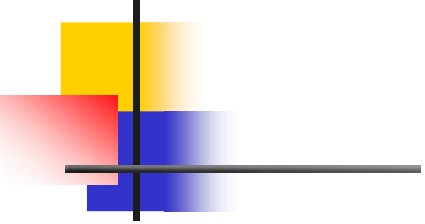
- Monitor all important functions

- Temperature
- Voltages
- Currents
- Sensors

NOTE: Data are usually multiplexed with a priority rating. There are usually two telemetry modes.

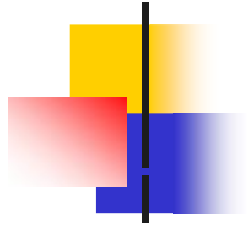


- Transmit data to earth
- Record data at TTC&M stations



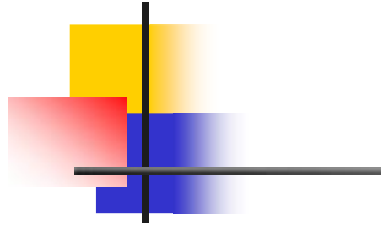
Telemetry Modes

- Non-earth pointing
 - During the launch phase
 - During “safe mode” operations when spacecraft loses tracking data
- Earth-pointing
 - During parts of the launch phase
 - During routine operations



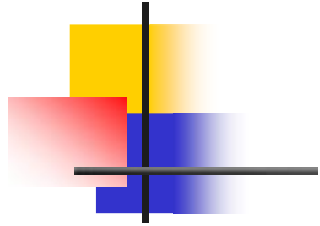
Tracking

- Measure range repeatedly
- Measure beacon Doppler or the communication channel
- Compute orbital elements
- Plan station-keeping maneuvers
- Communicate with main control station and users

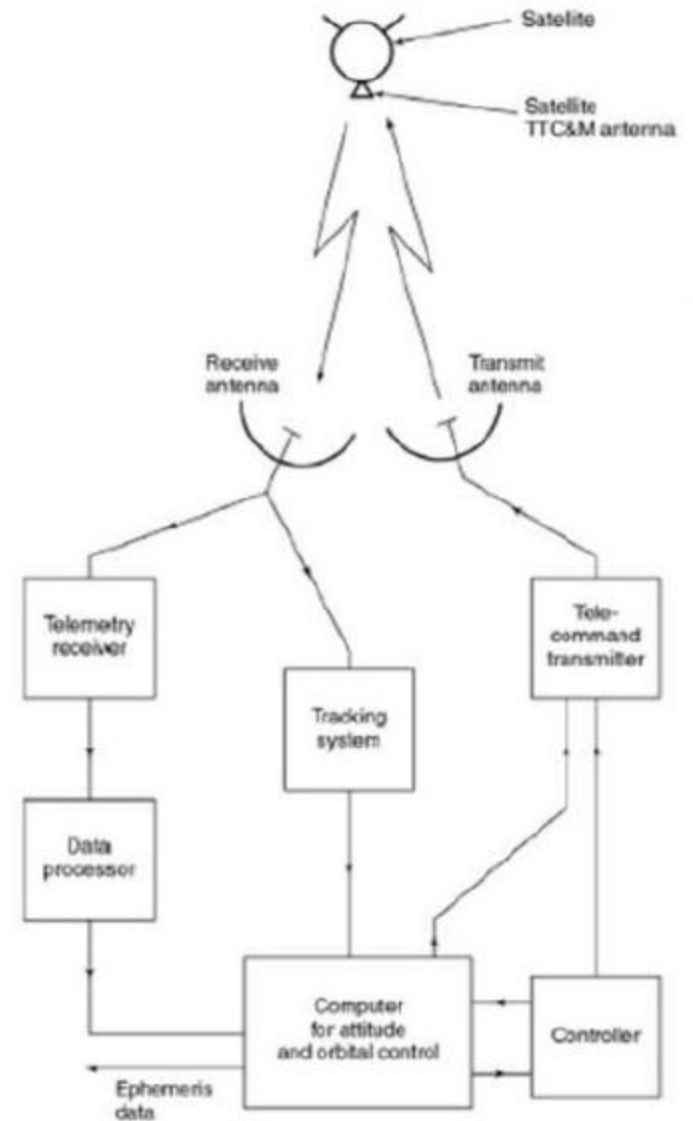


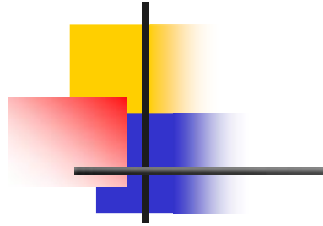
Command

- **During launch sequence**
 - Switch on power
 - Deploy antennas and solar panels
 - Point antennas to desired location
- **In orbit**
 - Maintain spacecraft thermal balance
 - Control payload, thrusters, etc.



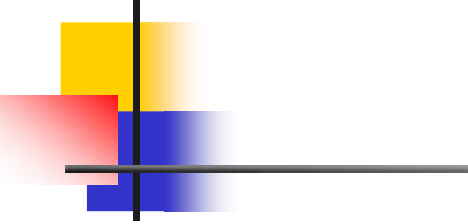
Typical TTC&M System





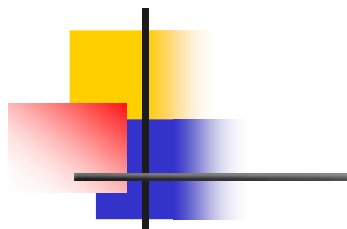
3. Power systems

- Solar cells
 - 1.39 kW/m² available from Sun
 - Cells 10 - 15% efficient (BOL = beginning of life)
 - Cells 7 - 10% efficient (EOL = end of life)
- Solar cell output falls when temperature rises
 - 2mV/°C
 - Three-axis is hotter (less efficient) than a spinner



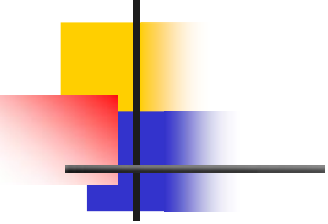
Backup Power System

- Batteries needed
 - During launch
 - During eclipse (< 70 minutes)
- Battery limits
 - NiCd 50% (DOD=depth of discharge)
 - NiH₂ 70% DOD



Power systems(Contd..)

- Batteries are “conditioned” before each eclipse season
 - Batteries discharged to limit
 - Batteries then recharged
- Typical NiH_2 battery can withstand 30,000 cycles (ample for GEO; would be 5 years in LEO)
- Typical battery voltages 20-50 V with 20-100 Ah capacity

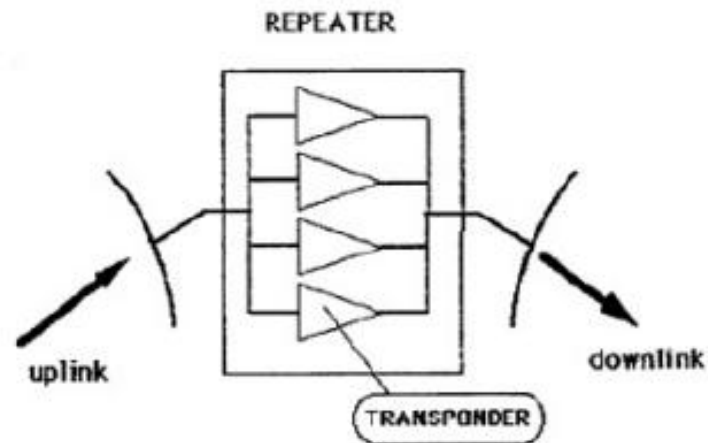


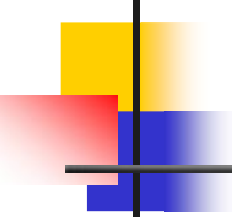
4. Communication subsystems

- Primary function of a communications satellite (all other subsystems are to support this one)
- Only source of revenue
- Design to maximize traffic capacity
- Downlink usually most critical (limited output power, limited antenna sizes)
- Early satellites were power limited
- Most satellites are now bandwidth limited

Repeaters and Transponders

- The total bandwidth (up to 500MHz-1.5GHz) is too large to be accommodated by a single amplifier.
- On the other hand, no carrier occupies such a bandwidth on its own.
- Hence the total REPEATER bandwidth is split into subbands (a few tens of MHz each). Each subband is amplified by a TRANSPONDER.

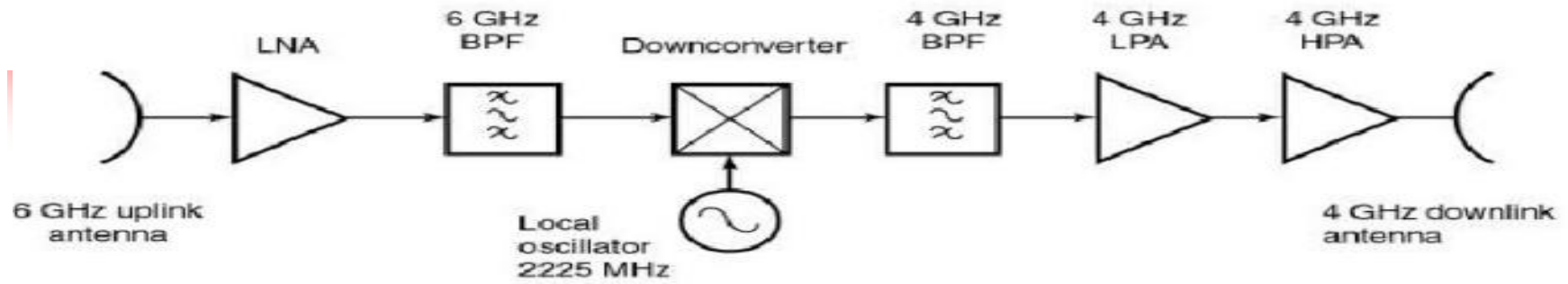




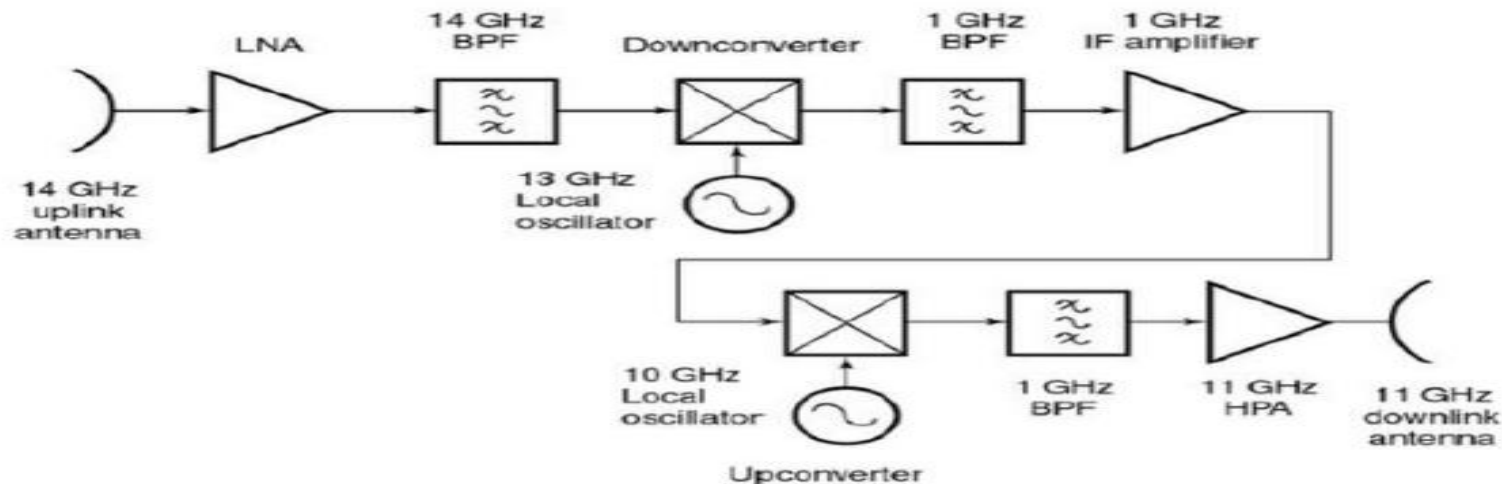
Types of payloads/Transponders

- Regenerative Transponders
 - Uplink is demodulated (IF=Baseband) and signal remodulated before transmission on downlink (not necessarily digital)
- On board Processing Transponders
 - Essentially digital in nature the uplink is demodulated and data recovered prior to processing. After processing, the data is reformatted for transmission on the downlink (store and forward transponders are an example)

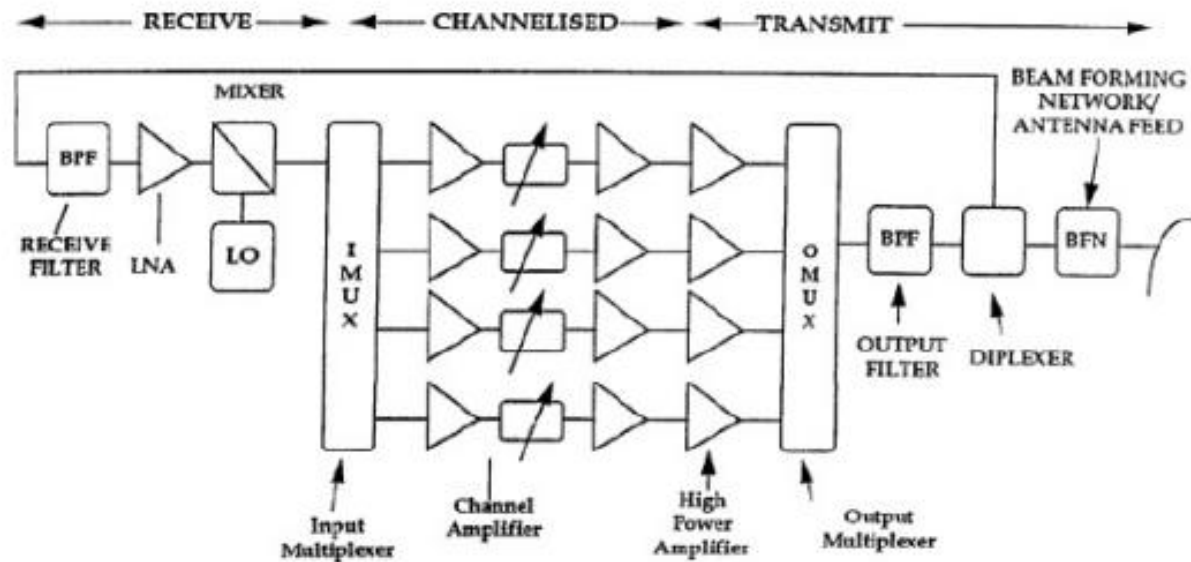
Single Conversion Bent pipe Transponder



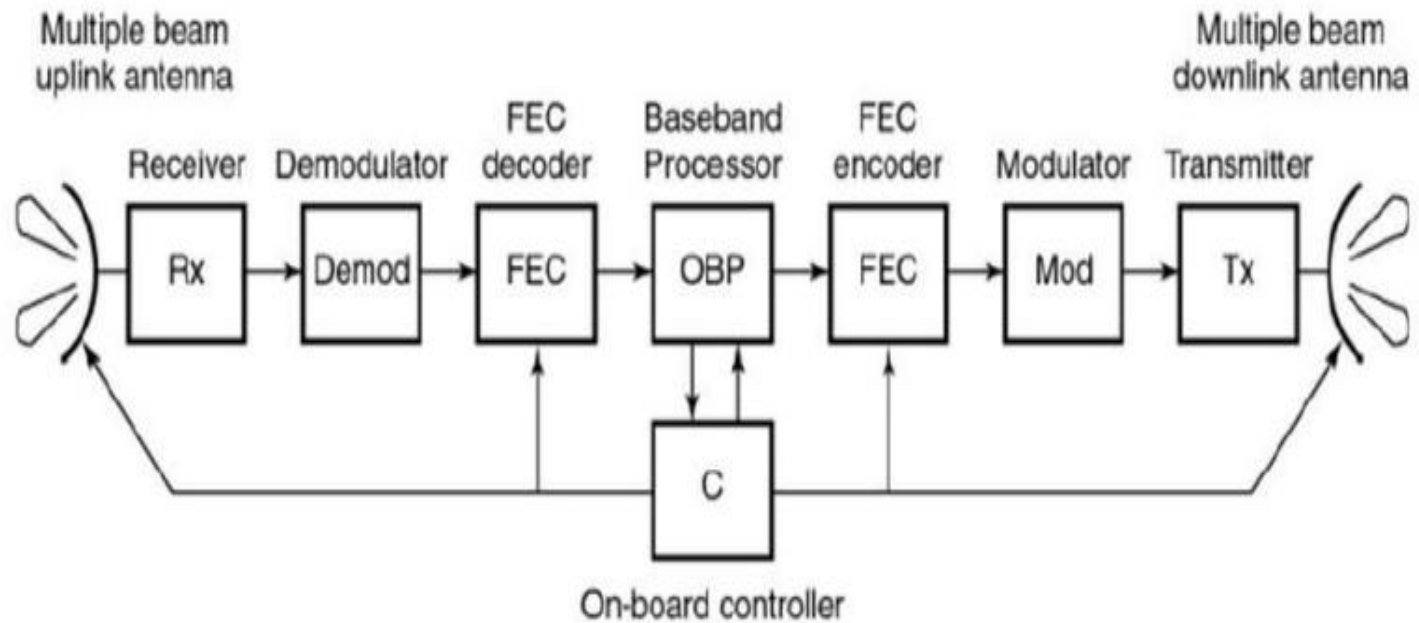
Double Conversion Bent pipe Transponder



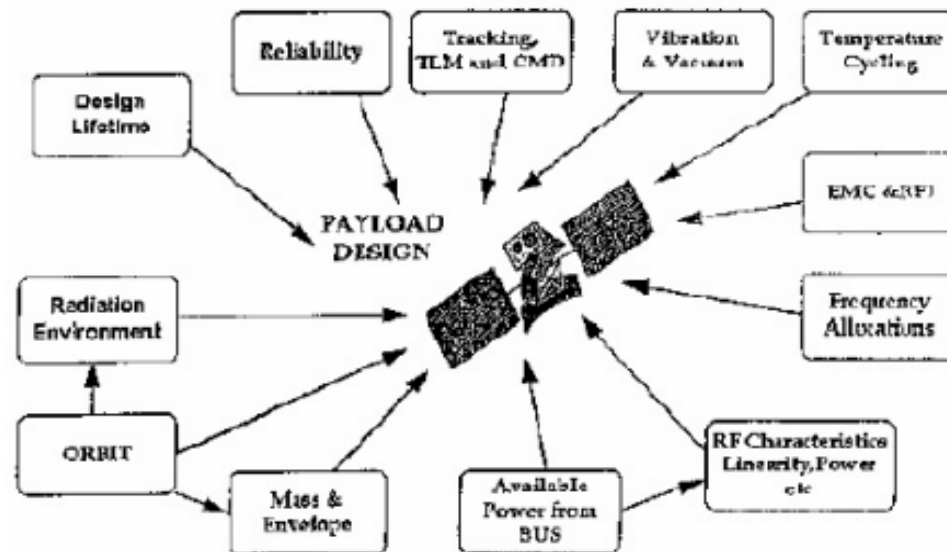
Basic Transponder elements



Onboard Processing Transponder



Factors affecting payload design





5. Satellite Antennas

Four main types of antennas are used on satellites

- **Wire antennas (Monopoles and Dipoles):** Primarily used at VHF and UHF to provide communication with the TTC&M system.
- **Horn antennas:** Used at microwave frequencies when relatively wide beams are required.
- **Reflector antennas:** Generally consist of Paraboloid reflector illuminated by one or more Horns and provide larger aperture than Horns.
- **Array antennas:** Used for generation of multiple spot scanning beams.

